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A Paleomagnetic Study of Upper Carboniferous Rocks from the Inner Sudetic Basin and the Bohemian Massif

K BIRKENMAJER, M KRS and A. E. M NAIRN

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A Paleomagnetic Study of Upper Carboniferous Rocks from the Inner Sudetic Basin and the Bohemian Massif

Abstract: A paleomagnetic investigation was carried out on oriented samples from two regions, the NE. and SW. parts of the Inner Sudetic basin, and from the Kladno-Rakovník and Plzeň basins some 200 km distant on the Bohemian platform. The directions of remanent magnetization from the two areas after magnetic cleaning (using both thermal and alternating field techniques) were significantly different. As the rocks examined can be shown to cover the same time interval, from Westphalian B/C to the end of Stephanian, it is suggested that the difference may be due to subsequent tectonic effects, and consideration of the results suggests that the Bohemian platform was the region affected. Geophysically the result suggests that tectonic effects may be an important cause of virtual pole dispersion; geologically it suggests the need for caution in paleomagnetic interpretation.

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INTRODUCTION

The work describes the results of studies on rocks of late Carboniferous age from the Inner Sudetic Basin in Lower Silesia and from the Kladno-Rakovník and Plzeň basins of Central Bohemia (Figs. 1 and 2). The results are relevant to the question of the Carboniferous-Permian boundary in a paleomagnetic connotation, and to the question of tectonic rotation in the interpretation of paleomagnetic data.

In recent years it has become common practice in some geological and geophysical literature to discuss virtual pole positions for geological epochs without much regard to the zonal position of the rocks within the epoch. Ideas of large-scale displacement have been formulated on the basis of significant differences between the virtual pole positions from rocks belonging to the same systems but on different continents. If such treatment of paleomagnetic data be accepted (and finer detail is seldom available), the nature of the gross smoothing involved in the use of a virtual pole position for a time period of the order of 50 m.y. is rarely made explicit. The smoothing operates in two ways. First, it may serve to conceal much of the fine detail containing information essential to an understanding of the large displacements proposed. It has yet to be shown whether the suggested movements were slow and continuous, or discontinuous and rapid. In this sense the often-quoted average rate of polar wandering of 1/3° per million years (Runcorn, 1956) can be extremely misleading. Second, in a spatial sense it is implicit

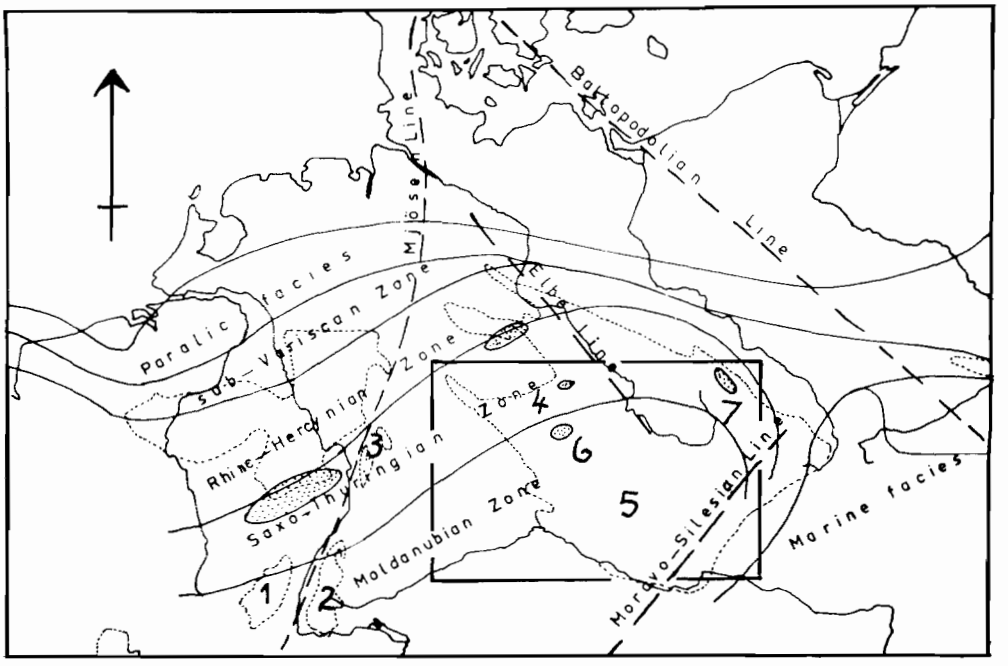


Figure 1. The zones of the Variscan Mountain System (*after* Brinkmann, 1954). Major tectonic lines are indicated *after* Stille (1949). Relative upstanding blocks are outlined by broken lines; the stippled areas represent Upper Carboniferous limnic basins. The main geological features of the inset zone are illustrated in Figure 2. Numbers refer to localities to which text reference is made.

1. Vosges
2. Schwarzwald
3. Odenwald
4. Krušné Hory
5. Bohemian Massif
6. Plzeň and Kladno basins (see also, Figs. 3 and 4)
7. Inner Sudetic basin (see also, Figs. 5 and 6)

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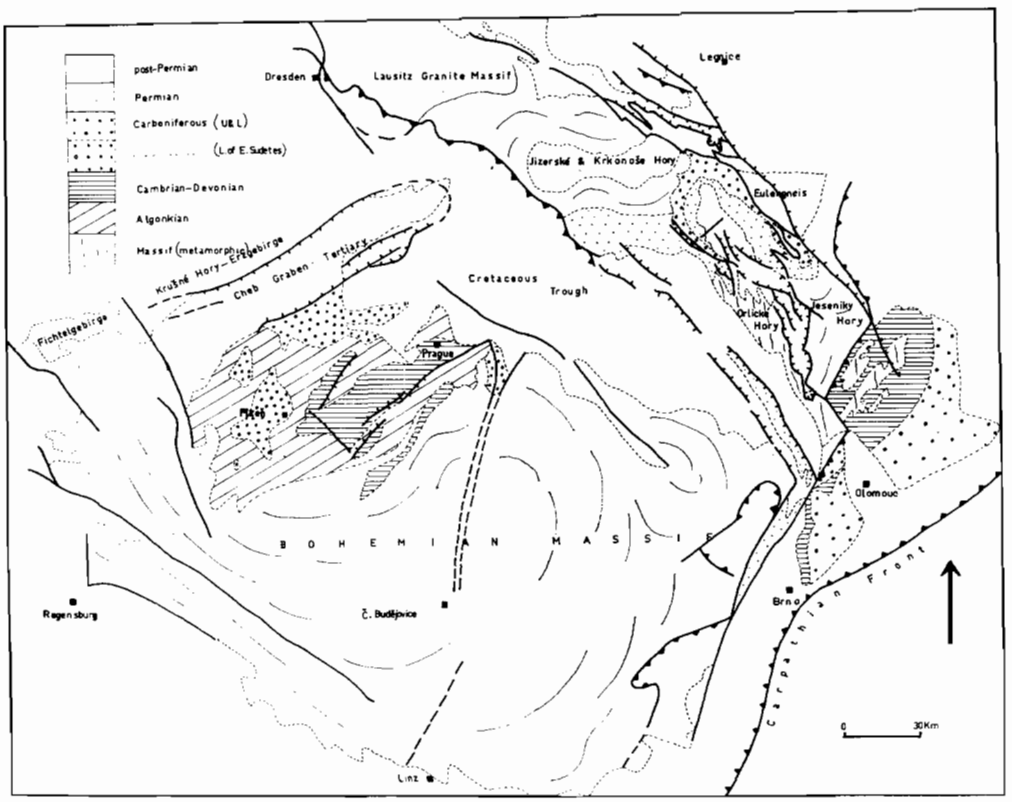


Figure 2. The main geological features of the Bohemian Massif and the Inner Sudetic Basin (simplified *after* Dorn, 1966). No attempt has been made to differentiate the various metamorphic and igneous rocks of the Massif.

in such comparisons that all points within the separate continental blocks remain fixed with respect to one another.

It is thus geologically and geophysically desirable to study in detail shorter time periods and smaller areas. With exact stratigraphical control the shorter time, although it must still be long in comparison with the assumed period of secular variation, may then give some indication of the way in which the directions recorded change with time. Near the boundaries between geological systems this would be of particular value, for as Irving and Parry (1963) have shown, the directions of magnetization of some Carboniferous rocks from Australia are similar to those found in Australian Permian rocks. To avoid confusion they went so far as to propose that an independent magnetic chronology be set up. The basins here studied are particularly apposite to the paleomagnetic study of the Carboniferous-Permian boundary, for there are lithologies suitable for paleomagnetic work, well dated because of the economic importance of the included coals. This time interval is not particularly well known paleomagnetically (see, however, Creer and others 1959; Everitt and Belshé, 1960; Nairn, 1960; Collinson and Nairn, 1960).

Until recently it has been generally assumed that relative movements within a continental block are unimportant, and one of the major problems has been to understand the dispersion of virtual pole positions found for some periods, such as the Carboniferous. The growing awareness of rotation of presumed tectonic origin, even within a continental area (Black, 1964; Kotásek and Krs, 1965; Ashworth and Nairn, 1965; Watkins, 1964; Irving and Opdyke, 1965; Kawai and others, 1962; van Hilten, 1962), necessitates the study of relatively restricted areas between which there are differences in tectonic development. This is the case here, for the Inner Sudetic basin belongs to the Saxo-Thuringian zone, and the coal fields of

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the Bohemian platform to the Moldanubian zone of Brinkmann (1954).

OUTLINE OF GEOLOGY

General Tectonic Pattern

The close of the Lower Carboniferous saw the retreat of the seas which had covered much of western and central Europe. This retreat followed orogenic movements which are referred to the Sudetic phase of the Hercynian orogeny. In central Europe the Variscan geosyncline was reduced to a narrow trough, leaving to the north a wide expanse on which coal forests developed, particularly during Westphalian times. It was as a result of these movements that the main outline of the Variscan mountains was established. A system of folds some 500 km broad narrowed eastward from Brittany and the Massif Central in France and stretched into central Germany, where the trend, in the region of the Elbe line (Brinkmann, 1954), turned sharply southeastward, passing through Silesia and Moravia into southwestern Poland (Fig. 1). This broad belt has been divided into four zones by Brinkmann (1954). These are, from south to north:

(1) The Moldanubian zone, including not only the Czech (Bohemian)[1] massif, but also the Schwarzwald (Black Forest) and Vosges. Within this zone, movements were restricted to block faulting, and sediments are thin or lacking. There was, however, considerable intrusive activity, with emplacement of granites, and some vulcanicity. Two of the coal basins here described, the Kladno and Plzeň (Pilsen) basins, belong to this zone.

(2) In the Saxo-Thuringian zone the crystalline basement appears at the surface in restricted areas such as the Odenwald and Krušné Hory (Erzgebirge). This was a region of heavy sedimentation during lower Paleozoic times and was much affected by Caledonian movements. After the Sudetic earth movements a number of intramontane basins developed, of which the Lower Silesian (Inner Sudetic) basin was one.

(3) The Rhine-Hercynian zone follows the line of the deepest region of the geosyncline and has a thick sequence of Devonian and Carboniferous (Culm) sediments. These are now generally found as open folds directed toward the northwest or northeast.

(4) The sub-Variscan foredeep, the marginal region of the ancient geosyncline bordered to the north by an epicontinental paralic zone, is a zone first folded in high Upper Carboniferous.

The main tectonic features of the region in which the coal basins studied occur are illustrated in Figure 2, the data for which are derived principally from Dorn (1966).

The Carboniferous of the Central Czech (Bohemian) Massif

On the Central Czech massif, Lower Carboniferous rocks are not known, and in the basins formed by simple block faulting apparently initiated during the Asturic phase, the earliest sediments date from the Westphalian B/C boundary. Three basins are known: Kladno-Rakovník, Plzeň, and Rosice-Český Brod (Rossitz-Böhmische Brod). The nearly flat-lying sediments are practically undisturbed except by faulting, more marked in the Plzeň basin than in the Kladno basin. This reflects renewed activity along older tectonic lines until as late as the Tertiary. Sedimentation occurred with only minor breaks (see Table 1) until the end of the Stephanian, with a maximum in Westphalian D and Stephanian B. The presence of Permian rocks has sometimes been assumed (for example, Dorn, 1966), but has not been proven (see Havlena, 1966).

The sedimentary rocks are predominantly continental sandstones and shales with economically important coal seams and some kaolin deposits. Volcanic rocks, porphyries, melaphyres and tuffs are known here as well as in the Inner Sudetic basin, and these may, as there, be related to weak movements of the Asturic phase. The flows may show kaolinization, often taken as an indication of climate. The succession forms a sequence of alternating red and gray beds. This alternation has been attributed to climatic variation, but may equally well be due to fluctuating ground-water level consequent upon minor uplifts or depressions of tectonic origin, or to sea-level changes which may themselves have a climatic cause.

The color alternation has been used to divide the succession into the sequence of four zones described below (*after* Havlena, 1966). The stratigraphical succession given by Havlena is shown in Table 1, alongside the succession given by Dorn (1966), which it supersedes because it is based on recent detailed studies.

Zone 1: lower gray zone of graywackes and gray mudstones with important coal seams. The flora of the Radnice coals is characteristic of

[1] For convenience, an attempt is made to give the German equivalents of some of the Czech and Polish names.

TABLE 1. THE UPPER CARBONIFEROUS STRATIGRAPHICAL SUCCESSION ON THE BOHEMIAN PLATFORM AND THE INNER SUDETIC BASIN*

		Inner Sudetic Basin				Bohemian Platform	
		Dorn 1966	Brinkmann 1954	Polish (NE) part Teissegre 1957 Oberc 1966	Czech (SW) part Tásler 1966	Havlena 1966	Dorn 1966
Autunian	U	L. Rotliegendes	L. Rotliegendes	L. Rotliegendes	Broumov Fm		Lihn Beds
	L				Chvaleč Fm / Bečkov mem / Vernerovice mem		
Stephanian	C	No C division given	Radowenz Beds	Stephanian formations	hiatus	U. Red Gp. (Klobouky Coals)	No C division given
	B	Radowanz Beds	Hexenstein Beds		Jívka mem Radvanice Coals / Odolov Fm	U. Gray Gp. (Kounov Coals)	Schlan Beds
		Hexenstein Arkose					Teinlitz Beds
	A	Idastall Beds	Idastall Beds		Svatonovice mem Z. Nejedly Coals	Lr. Red Gp. (Nevřeň Coals)	Kladno-Pilsen-Radnice Beds
Westphalian	D	Schadowitz Beds	Schadowitz Beds	? hiatus	? hiatus	Lr. Gray Gp. (Nyrany Coals) V	Lower Gray Beds
	C	Zdiarek Beds		Žacléř Fm	Vilemína V Coals V / Žacléř Fm Markoušovice V Coals V	Radnice Coals V	Arkose, Kladno Coals Tuff
	B	Schatzlar Beds	Schatzlar Beds		(Jan Šverma Coals)		Radnice Coals
	A	Weisstein Beds					
Namurian	C	Waldenburg Beds	Weisstein Beds	Biały Kamien Fm	Biały Kamien Fm		
	B		hiatus	hiatus			
	A		Waldenburg Beds	Wałbrzych Fm	Wałbrych Fm		

(After Tásler, 1966; Brinkmann, 1954; Oberc, 1966; Teisseyre, 1957) Some of the important coal and volcanic horizons (V) are indicated.

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Westphalian C, close to the Westphalian B/C boundary. In the absence of fossils, the upper limit of the zone is defined on lithological grounds.

Zone 2: lower red zone of sandstones and mudstones which transgress over the lower gray zone onto the basement. Havlena (1966) describes the flora as characteristic of uppermost Westphalian D and lower Stephanian. *Walchia* and *Mixoneura* are reported; a similar floral association is known in the Inner Sudetic basin.

Zone 3: upper gray zone with a flora reliably identified as Stephanian B/C toward the top. This floral assemblage of the Kounov coal series, with *Annularia spicata* (Gutb.), *A. stellata* (Schloth.), *A. sphenophylloides* (Zenker), *Sphenophyllum oblongifolium* (Germ. and Kaulf.), *Sigillaria brardi* (Brong.), *Pecopteria feminaeformis* (Schloth.), *Odontopteris subcrenulata* (Rost), *Callipteris triguim* (Franke), and *Alethopteris bohemica* (Franke), permits correlation with the Radvanice coals of the Jívka member of the Odolov Formation in the Inner Sudetic basin.

Zone 4: upper red zone of red sandstones and pelitic sediments. These rocks, formerly considered Permian, contain a flora associated with the Klobouky coals in which *Sphenophyllum thoni* (Mohr), *Asolanus camptotaenia* (Wood), *Pecopteria pernica* (Němejc), *Odontopteris genuina* (Grand-Eury), *Callipteridium* sp., and *Alethopteris bohemica* (Franke) have been identified, and the whole series is regarded as still belonging to the Stephanian (Havlena, 1966).

The Carboniferous of the Inner Sudetic Basin

The Inner Sudetic basin is a narrow, elongate trough lying between crystalline massifs. It is bounded to the southeast by the Jeseníky Mountains (Schneegebirge) and to the northwest by the Jizerské Hory and Karkonosze (Krkonoše or Riesengebirge), and flanked to the northeast and southwest by the Sowie Góry Gneiss massif (Eulengneiss) and the Orlické Hory (Orlickie Góry or Adlergebirge), respectively. Major northwest-southeast-trending faults paralleling the Elbe line which help define the basin, a primarily Variscan structure, were reactivated in later periods, during which time overthrusting developed.

From Namurian times on into Lower Permian times there was more or less continuous sedimentation in the Inner Sudetic basin. The major break occurred after the Middle Rotliegendes, above which there are important conglomerates (fanglomerates). There is a certain amount of disagreement among various authors over the exact stratigraphy, and in Table 1 an attempt has been made to correlate the succession given by Brinkmann (1954) and Dorn (1966) with the most recent work summarized by Tásler (1966) for Czechoslovakia, and by Teisseyre (1957) and Oberc (1966) for Poland. Asturic phase movements were weak, although the volcanic activity accompanying granite emplacement is related to them. Further vulcanicity occurred in the Middle Rotliegendes. Most of the stratigraphic divisions show a tendency to begin with coarse-grained, even conglomeratic, sediments and trend toward finer-grained beds.

The lowermost formation, the Wałbrzych (Waldenburg) Formation, is confined to the Polish (northern) part of the basin and represents Namurian A. It consists of some 200 m of quartz conglomerates, sandstones, siltstones, and some coals. The Sudetic phase was weak here, and the Wałbrzych Formation rests conformably upon the Culm. It is unconformably overlain by the Bialy Kamień (Weisstein) Formation. This formation also reaches its maximum thickness of 300 m in the northern part of the Inner Sudetic basin, and represents the time interval Namurian C/ Westphalian A. It is made up of conglomerates, sandstones, and siltstones. In the southeastern part of the basin the whole of the Namurian and Westphalian A are represented by the Nowa Ruda Formation. The Nowa Ruda Formation does not appear to be conglomeratic, being made up of clayshales, kaolinitic shales, siltstones, and some coals. There is a basal regolithic breccia (Teisseyre, 1957; Oberc, 1966).

In the southwest the Žacléř (Schatzlar) Formation is transgressive and rests directly upon crystalline rocks. Three sedimentary cycles have been recognized in a sequence of some 600 to 700 m of conglomerates, sandstones, and shales with coals. The floral grounds suggest the lowest cycle covers the time range Westphalian A to low Westphalian B. In both the middle (Westphalian B) and the upper (Westphalian C) divisions the earliest rocks are conglomerates. In both too, there are volcanics, usually in the form of effusive sheets. Coals are less abundant than in the lower division, and the floral evidence of age is obtained from the Markoušovice coals and the seams worked at the Vilemína pit. The Asturic vulcanicity in the northern part of the basin is referred to the Westphalian B–C (Grocholski, 1962).

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The succeeding Odolov Formation, divisible into a lower Svatoňovice and an upper Jívka member, shows marked lithological differences from the underlying Žacléř Formation, differences attributed to climatic change (?) and new sources of sediments which resulted from Asturic phase movements. Both members have coarse sandy, even conglomeratic, basal beds. The lower members have been dated as Westphalian D to lower Stephanian from the examination of coals from the Z. Nejedly pit, whereas the Radvanice coals of the Jívka member have been correlated with the Kounov coals of Central Bohemia.

According to Tásler (1966), a break in deposition, with some erosion of the uppermost beds of the Jívka member, intervened before deposition of the Chvaleč Formation, the lower Rotliegendes referred to by both Brinkmann and Dorn. It has been divided into two stages, both beginning with conglomerates and passing to finer-grained beds. In the lower Vernéřovice member a thin coal, the "*Walchia*" seam, is developed. Toward the base of the upper Bečkov member there are important tuff, porphyry and melaphyre eruptives.

In the Polish part of the basin the 480 to 650-m-thick Stephanian beds are developed as conglomerates, sandstones, and siltstones, with some reworked volcanic material from the Asturic volcanics. Locally, a hiatus can be recognized at the base of the Stephanian. The overlying Rotliegendes, which are up to 2000 m thick, show some of the features of a cyclic molasse-type sedimentation (Dziedzic, 1961). Such cycles usually begin with conglomerates (fanglomerates) and grade upward into sandstones and shale. There are two cycles in the lower Rotliegendes (Autunian), one in the middle Rotliegendes and one in the upper Rotliegendes. An eruptive complex of both acidic and alkalic volcanics is referred to the close of the third cycle (Dziedzic, 1961; Kozłowski, 1963).

An indication of the localities sampled is given on the geological sketch maps of the three coal basin studies (Figs. 3, 4, 5, and 6). Natural outcrops in the Kladno-Rakovník and

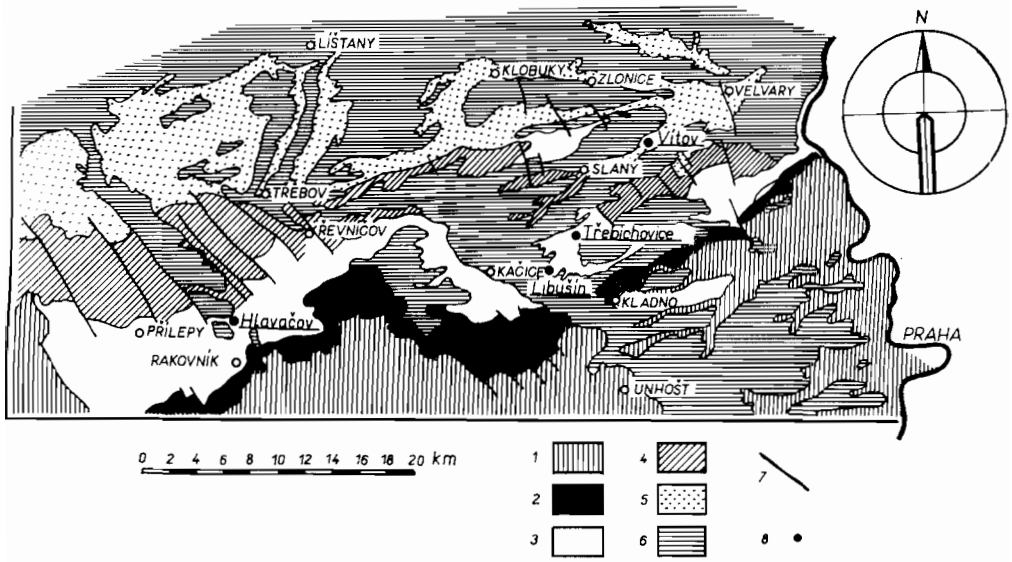


Figure 3. Geological sketch map of the Kladno-Rakovník coal basin.

1. pre-Carboniferous rocks
2. Lower Gray Zone, Upper Carboniferous
3. Lower Red Zone, Upper Carboniferous
4. Upper Gray Zone, Upper Carboniferous
5. Upper Red Zone, Upper Carboniferous
6. post-Carboniferous rocks
7. fault lines
8. sampling localities

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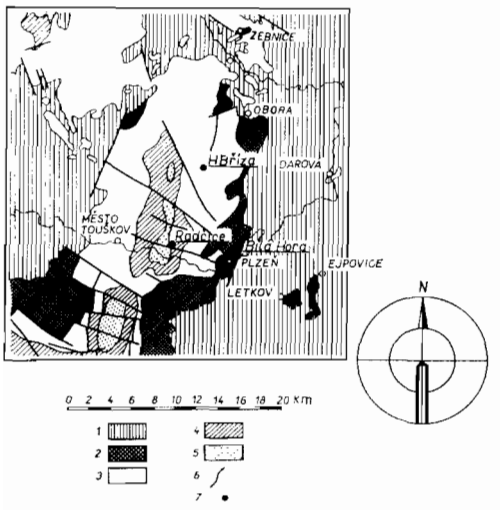


Figure 4. Geological sketch map of part of the Plzeň coal basin.

1. pre-Carboniferous rocks
2. Lower Gray Zone, Upper Carboniferous
3. Lower Red Zone, Upper Carboniferous
4. Upper Gray Zone, Upper Carboniferous
5. Upper Red Zone, Upper Carboniferous
6. fault lines
7. sampling localities

Plzeň basins are generally lacking, and sampling was consequently restricted. In the Kladno-Rakovník basin (Fig. 3), two of the four sites were in extensive open cast workings in rock of the lower red zone. Samples of red sediments in the lower gray zone were obtained from the Libušin mine shaft extensions. At the only natural outcrop visited, samples were obtained from the upper red zone. In the Plzeň basin (Fig. 4) this same lack of natural outcrops made it possible to sample only three sites covering zones 1 and 2, and these were all in mine workings. Surface outcrops occur more frequently in the Inner Sudetic basin (Figs. 5 and 6), and most of the volcanic rocks were collected in quarries. The sediments were collected from both natural and artificial profiles.

In collecting from igneous rocks, an attempt was made to obtain a minimum of six samples from each site, with the spacing dependent upon the possibilities at each outcrop. In the case of the sedimentary rocks one sample per bed was usually collected and as many different units as possible sampled. In the case of mine workings or open cast sites this number was sometimes considerable (the maximum being 30). Minor faulting was not everywhere easy to detect, and concealed areas may have resulted in some horizons being duplicated; although no accurate estimate can be given of this, it is unlikely to have exceeded 10 percent.

PALEOMAGNETIC OBSERVATIONS

It is only possible to consider the problem of dispersion and study possible tectonic effects if secondary magnetizations are not present or have been removed from the rocks collected, and if enough horizons have been sampled in all the regions concerned to give reasonable assurance that secular variation will bias none of the resultant mean directions of magnetization. These requirements rest upon the dual assumptions that the hard magnetic component is original, dating from the time of formation of the rock or rocks concerned, and that the effects of secular variation can be averaged out. The latter point was first brought out by Hospers (1954), and later by Runcorn (1959), and seems to be borne out by more

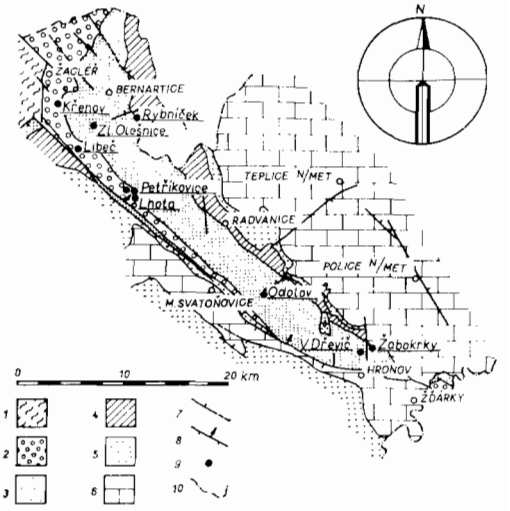


Figure 5. Geological sketch map of the southwest part of the Inner Sudetic coal basin.

1. Crystalline rocks
2. Žacléř Beds, Westphalian B/C
3. Odolov Beds, Westphalian D—Stephanian
4. Autunian (lower Rotliegendes)
5. Saxonian (middle and upper Rotliegendes)
6. Upper Cretaceous
7. fault lines
8. Hronov-Poříčí shear zone
9. sampling localities
10. Czechoslovak-Poland frontier

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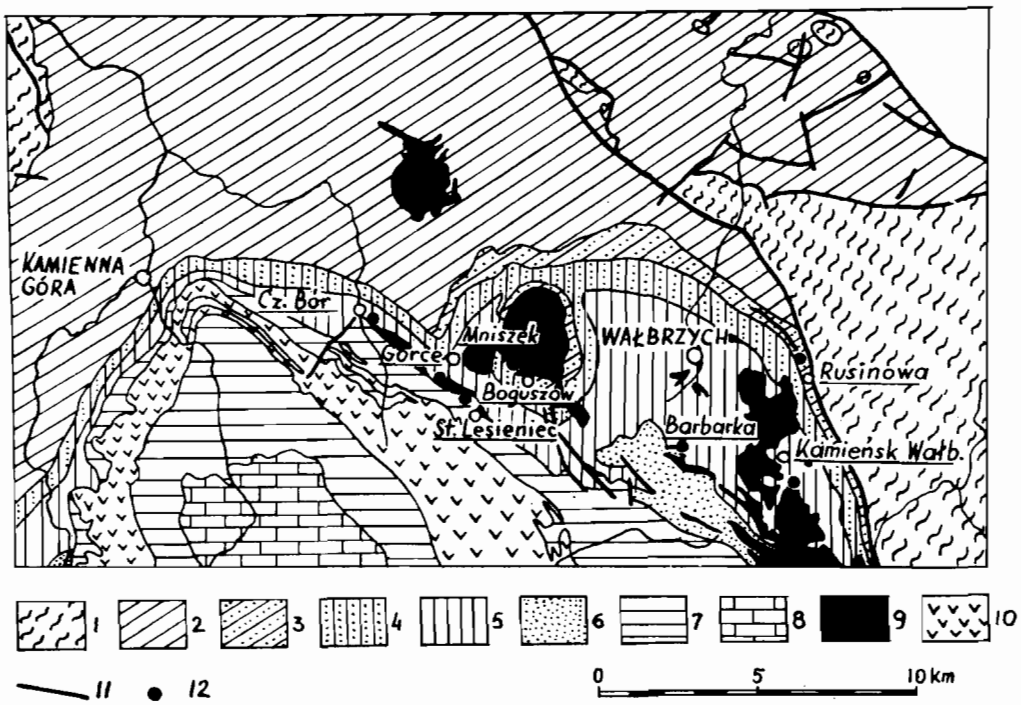


Figure 6. Geological sketch map of the northern part of the Inner Sudetic basin.

1. Metamorphic rocks
2. Upper Devonian and Culm
3. Wałbrzych Formation, Namurian A
4. Biały Kamien Formation, Namurian C and Westphalian A
5. Žacléř Formation, Westphalian A–C
6. Westphalian D and Stephanian, not subdivided
7. Rotliegendes, sedimentary rocks
8. Post-Rotliegendes rocks/Zechstein, Bunter, Cretaceous
9. Vulcanites effusive and intrusive, mostly porphyries, principally Westphalian B–C
10. Vulcanites, mostly effusive/porphyries, melaphyres, etc./middle Rotliegendes
11. Faults and overthrusts
12. Sampling localities

recent detailed work on Tertiary and Recent rocks (Irving, 1964). The first assumption requires both data on the magnetic properties of the rocks from paleomagnetic study, and a consideration of the geology, which will be considered in the following section. Fortunately, in the areas considered here there is a variety of lithologies known by experience to be suitable for paleomagnetic studies; red sandstones and mudstones, tuffs, melaphyres and dolerites give a reasonable chance of meeting these conditions. Although not all lithologies are present at all sites, where different rock types occur the results are consistent (see Table 2a–2d), and of the rocks sampled, only the tuffs from the Kladno-Rakovník and Plzeň basins proved totally unstable.

Two types of instruments were used to measure the directions of remanent magnetization: rapid determinations were made with a low-sensitivity short-period astatic magnetometer when the intensity was sufficiently high (of the order 10^{-4} emu/cc), while rocks with lower intensities of magnetization were measured using a rock generator magnetometer (Jelínek, 1966). All rocks were subjected to progressive demagnetization, both alternating field and thermal demagnetization techniques having been applied. In general, alternating fields proved satisfactory in dealing with igneous rocks, and thermal techniques in dealing with sedimentary rocks.

Alternating field demagnetization methods have become fairly standard since the work of

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TABLE 2.

Locality	Stratigraphy	Declination	Inclination
(a) *The Kladno-Rakovník basin*			
Libusín	Zone 1 Westphalian C red sandstones	208.8	− 8.4
Vitov	Zone 4 Stephanian red mudstone	208.9	+ 1.4
Trebichovice	Zone 2 Stephanian red mudstone	210.8	− 3.9
Hlavacov	Zone 2 Stephanian red mudstone	213.2	− 4.9
Mean value		211.7	− 5.4
(b) *The Plzeň basin*			
Radcice	Zone 2 Stephanian red mudstone	211.7	+ 4.7
Horní Bríza	Zone 2 Stephanian red sandstone	211.3	−15.0
Bílá Hora	Zone 1 Westphalian C red sandstone	219.2	+19.8
Mean value		214.7	+ 2.1
(c) *The Inner Sudetic basin (N.E. Bohemia)*			
Krenov	Westphalian B. Žacléř beds porphyry tuffs	190.5	− 5.1
Velký Drevic	"Stephanian" graywacke	212.3	+33.1
Odolov	"Stephanian" sandstone	206.4	− 9.3
Lhota	"Upper Carboniferous" sandstone	191.7	− 1.6
Petrikovice	Westphalian C Žacléř beds sandstone	188.0	+ 5.2
Zlata Olesnice	Westphalian D–Stephanian sandstone	181.4	+11.7
Rybnicek	"Upper Carboniferous" sandstone	207.3	− 6.3
Zaborkrky	"Stephanian" sandstone	201.8	+16.8
Petrikovice	Westphalian C–D(?) sandstone	187.8	−15.5
Libec	Westphalian B melaphyre	192.6	+18.3
Mean Value		196.2	+ 2.2
(d) *The Inner Sudetic basin (Lower Silesia)*			
Płoczki Gorne*	post-Westphalian pre-lower Rotliengendes sill	196	− 2
Kamieńsk Waldbrzyski	Westphalian B/C porphyry neck	189	− 4
Rusinowa	post-Namurian A probably high Westphalian sill	214	−15
Mount Barbarka	Žacléř beds Westphalian B/C porphyry extrusive	182	−46
Boguszów	post-Westphalian B probably Westphalian B/C felsite porphyry	194	− 3
Mount Mniszek	post-Westphalian B probably Westphalian B/C felsite porphyry	184	−21
Gorce	post-Westphalian B pre-L. Rotliegendes felsite porphyry extrusive	198	+10
Stary Lesieniec	post-Westphalian B pre-L. Rotliegendes extrusive	254	−64
Czarny Bór	post-Westphalian B pre-L. Rotliegendes extrusion	205	−13
Mean Value		196	−12
Mean Value	for the Inner Sudetic basin excluding sites where $\alpha > 20°$	195.1	−1.8

The site mean directions of magnetization recorded from the Kladno, Plzeň and Inner Sudetic basin (Bohemia) represent the mean from the number of horizons (1 sample per horizon) indicated in column N. The results in table 2d from the Inner Sudetic basin (lower Silesia) represent the mean for a single horizon based on the number of samples given in column N.

* This locality lies in the North Sudetic basin.

Creer (1959), Irving and others (1961), and As and Zijderveld (1958). The method used in this study was to rotate samples simultaneously about two axes in the demagnetizing solenoid. Field free space, controlled to better than 50 gamma, was maintained while the alternating magnetic field was applied. Smooth field reduction was ensured by using a motor to power the extraction of the movable electrode of a potential divider. In some cases demagnetization was continued in fields of up to 600 gauss, but in general the igneous rocks were cleaned by fields of less than 200 gauss. Some typical demagnetization curves are shown in Figure 7, a and b. Sedimentary rocks were little affected by alternating fields, even up to the highest fields employed (1400 gauss).

Thermal demagnetization of the sedimentary rocks was carried out in a non-inductively wound furnace with a higher degree of field cancellation than used in a.c. demagnetization experiments. Thermal demagnetization curves are shown in Figure 8, a and b. It was found that in most of the sediments the best results were obtained at a temperature of 600° C. Changes in the direction of magnetization during the course of demagnetization are illustrated in Figure 8. The resulting mean direc-

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PALEOMAGNETIC DATA

α	K	R	N	Treatment
5.2	34.9	22.37	23	NRM given, no sig. difference after cleaning at 600°
6.1	230.4	3.99	4	Cleaned at 600°
6.6	30.5	16.48	17	Cleaned at 600°
3.3	66.1	28.58	29	Cleaned at 600°
2.6	42.0	71.29	73	All measurements
12.4	24.6	6.76	7	Cleaned at 600°
4.6	33.6	29.14	30	Cleaned at 600°
5.7	23.9	26.87	28	Cleaned at 600°
5.1	12.9	60.06	65	All measurements
4.6	84.1	12.86	13	A.C. field 500 gauss
28.3	8.3	4.52	5	Large scatter not reduced A.C. fields of 1200 gauss
5.4	66.2	11.83	12	A.C. fields of 1000 gauss, thermal cleaning 600°
4.2	99.1	12.88	13	A.C. fields of 1000 gauss, thermal cleaning 600°
9.2	54.4	5.91	6	A.C. fields of 1000 gauss, thermal cleaning 600°
23.8	27.9	2.93	3	A.C. fields of 1000 gauss, thermal cleaning 600°
34.7	4.7	4.93	6	A.C. fields of 1000 gauss, thermal cleaning 600°
13.0	9.0	14.34	16	Cleaned at 500°
5.5	88.2	8.91	9	A.C. fields of 1000 gauss, thermal cleaning 600°
23.2	11.8	4.66	5	A.C. field of 250 gauss
4.5	12.0	80.76	88	
13.9	44.4	3.93	4	A.C. field of 170 gauss
18.5	18.1	4.78	5	Demagnetization in A.C. fields, minimum scatter in NRM
24.8	25.7	2.92	3	A.C. fields of 255 gauss
11.6	34.4	5.85	6	A.C. fields of 255 gauss
6.1	228.2	3.99	4	A.C. fields of 85 gauss
19.1	24.2	3.88	4	Demagnetized in A.C. fields, minimum scatter in NRM
6.7	191.9	3.98	4	A.C. fields of 255 gauss
17.6	19.8	4.80	5	Demagnetized in A.C. fields, minimum scatter in NRM
6.7	340.8	2.99	3	A.C. fields of 255 gauss
13.6	17.4	7.60	8	Results from Stary Lesieniec excluded
3.99	17.61	..	76	

α = radius of the circle of confidence (95%).
K = parameter giving a measure of dispersion (Fisher, 1953)
R = length of vector mean
N = number of samples.

tions and various other statistics are shown in Table 2a to 2d, and the mean directions are illustrated in Figure 9a to 9c. Regional mean directions of magnetization and the computed virtual pole positions with corresponding confidence limits are given in Table 3.

In computing results, minimum site dispersion of igneous rocks subjected to a.c. cleaning was the criterion used in selecting the direction considered as best representing the original direction of magnetization. This was based on the assumption, common in rock magnetism, that at a given site in an igneous rock unit such as a lava flow, the time of cooling through the Curie temperature during which time a thermo-remanent moment was acquired, is measurable in years. In the sedimentary rocks, the bed was used as a recognizable unit on the assumption that this, too, was probably the record of a single event of short time duration. Analyzing the change in direction during thermal demagnetization using the technique of As and Zijderveld (1958) permits recognition of the stage at which secondary magnetization effects have been removed. This was found to be at a temperature of 600° C in most cases (Figure 8), and as this coincided with the minimum site

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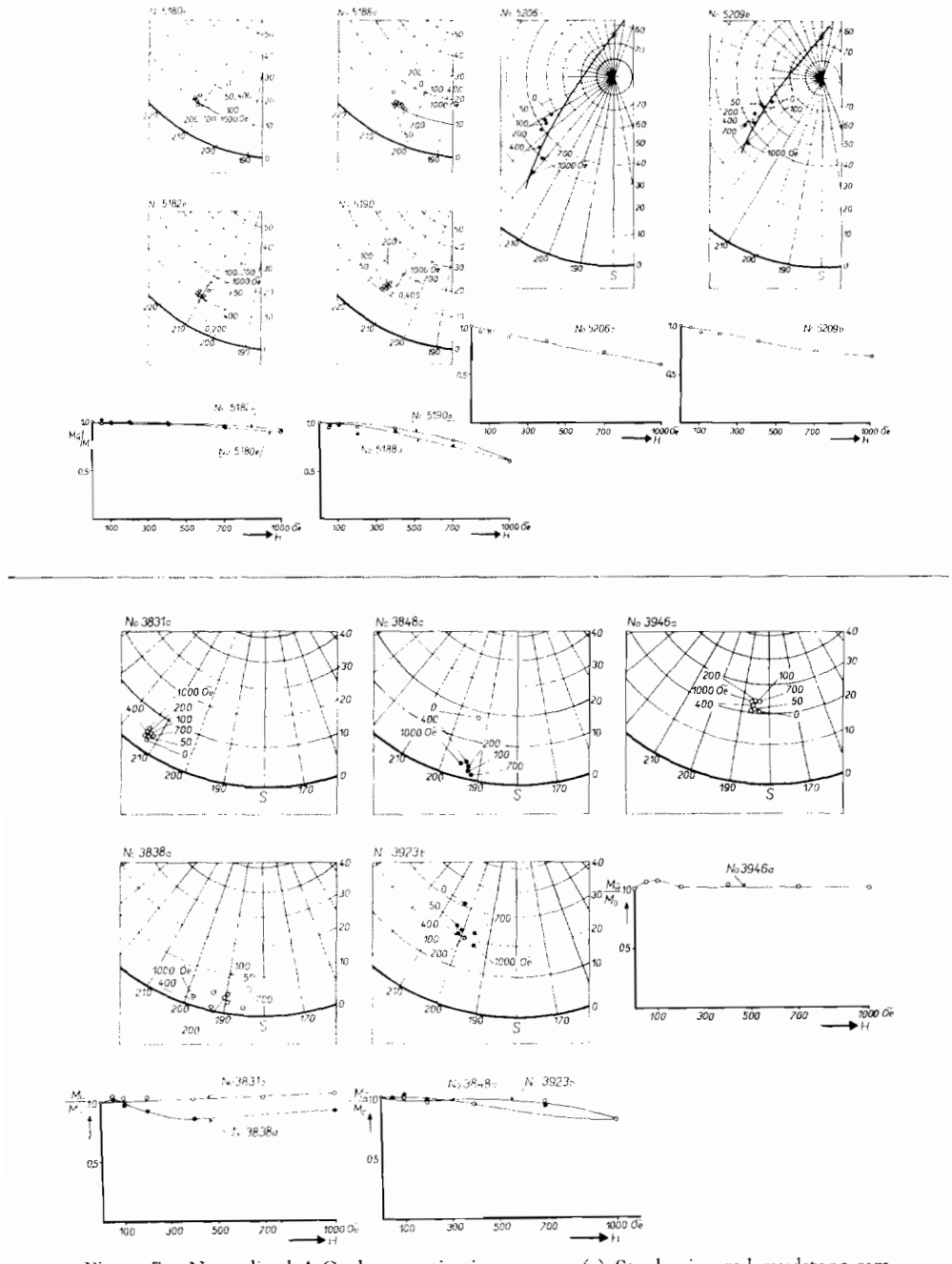


Figure 7. Normalized A.C. demagnetization curves: (a) Stephanian red mudstone samples from three sites in the Kladno basin (5180e, 5182e, Hlavačov, 5188a, 5190a Trebichovice, 5206a, 5209b Vítov); and (b) Westphalian and Stephanian samples from four sites in the Inner Sudetic basin (3838a Lhota, 3848a Petříkovice, 3923b Žaborkrky, 3946a Petříkovice valley). The changes in the direction of magnetization during demagnetization are shown on the accompanying stereograms (○ north-seeking pole pointing upward, ● north-seeking pole pointing downward).

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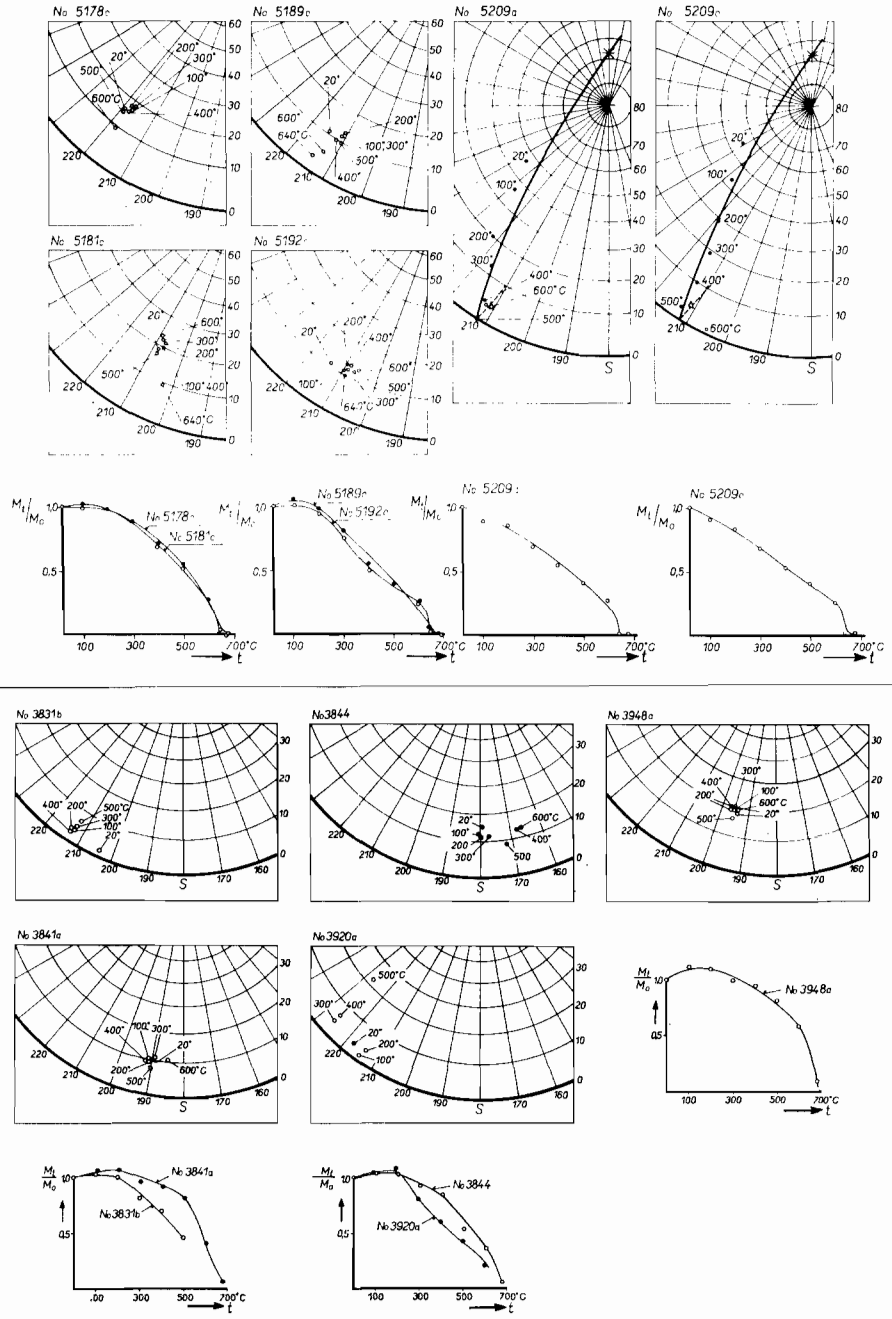


Figure 8. Thermal demagnetization curves of: (a) Stephanian red mudstone samples from three sites in the Kladno basin (5178c, 5181c Hlavačov, 5189a, 5192c Třebichovice, 5209a, 5209c Vítov); and (b) Westphalian and Stephanian from five sites in the Inner Sudetic basin (3831b Odolov, 3841a Lhota, 3844 Petříkovice, 3920a Žabokrky, 3984a Petříkovice valley). The changes in the direction of magnetization during the course of demagnetization are shown in the accompanying stereograms. These diagrams should be compared with the results shown in Figure 7, especially for the Vitov site.

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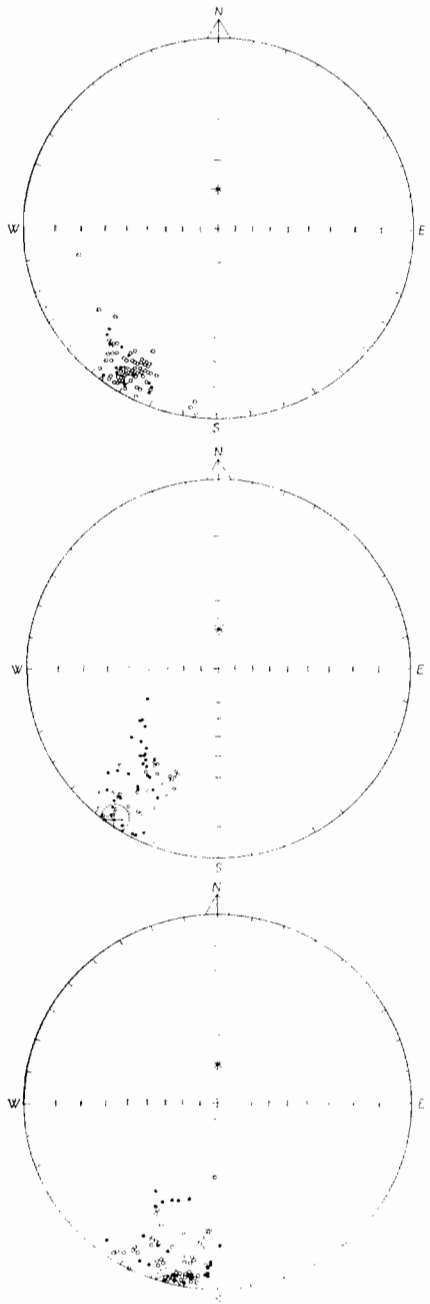


Figure 9. Stereograms of the directions of magnetization after magnetic cleaning of: (a) Kladno-Rakovník basin; (b) Plzeň basin; (c) southwest (Czech) and the northeast (Polish) parts of the Inner Sudetic basin. Each point represents a single horizon. ● north-seeking pole pointing downward, ○ north-seeking pole pointing upward, -⊙- mean direction of magnetization enclosed by its circle of confidence, * theoretical dipole field direction.

dispersion, the mean direction at this stage was adopted. The number of stages in thermal demagnetization required by the As and Zijderveld method was then reduced. One consequence of this technique is that the site mean directions in sedimentary rocks are not strictly comparable either with one another or with igneous sites, each being based upon a variable number of horizons. The means in Table 2a to 2d are therefore to be regarded as a good but qualitative indication of the general agreement of directions of magnetization among sites within each basin (Fig. 10). The mean direction for each basin was computed by giving each bed or igneous horizon unit weight. Where a site dispersion has confidence limits above 20°, all samples from that site have been arbitrarily excluded from the computation of the mean direction of magnetization of the coal basin. This may not be strictly justified, for only a few horizons are concerned, and therefore secular variation effects may not be negligible; however, the mean direction in which such samples are included does not differ significantly from the basin means given in Table 2a to 2d.

DISCUSSION

The Magnetization of the Carboniferous Rocks of the Central Bohemian Massif and the Inner Sudetic Basin

Within each basin the directions of magnetization of the rocks show a high degree of consistency after the removal of secondary magnetization. There is complete accord between the results of measurements of igneous and sedimentary rocks (Fig. 10). Yet the results from the Inner Sudetic basin, which do not differ significantly from the directions of the succeeding Permian beds, both lower and middle Rotliegendes, are significantly different from the results from the Central Bohemian massif less than 200 km distant.

As direct correlation of some of the coal horizons is possible, there can be no question of the difference being a result of an age difference: if the time span in the Sudetic region is greater, the difference is small, and the number of samples from the time not represented in

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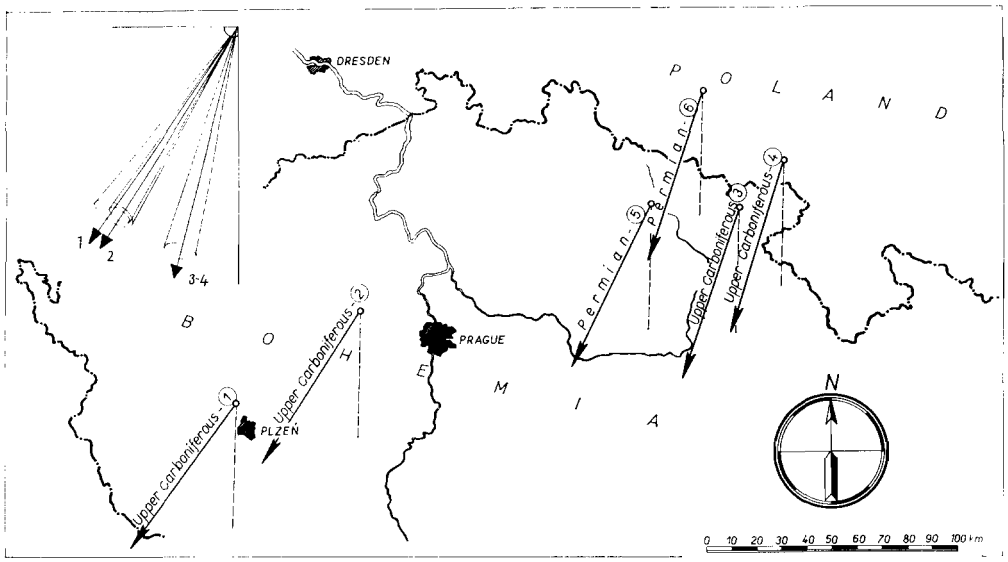


Figure 10. The mean magnetic declinations from different sites. For reference, the Permian direction and the meridian are drawn. In the upper right corner the mean directions enclosed within their circles of confidence are assembled: (1) Plzeň (2) Kladno (3) Inner Sudetic basin combined result. This illustrates the significant difference in mean direction of magnetization between the Sudetic basin and the Bohemian platform.

Bohemia is insufficient to bias the mean. The difference can scarcely be attributed to the methods or the materials used, since the same techniques were applied equally to rocks from all areas, and the material examined had not been subjected to recent weathering, having been collected primarily from quarries and mine workings. Nor can it be the result of the complete remagnetization of the rocks of one area. The latter would imply, on magnetic grounds, either complete reheating above the Curie temperature, which is geologically ruled out by the condition of the coals, or remagnetization of the whole series of rocks as a result of oxidation processes. It would then have to be assumed that the processes worked preferentially in a sequence of beds over 6000 feet thick without leaving a trace in interbedded porous nonred horizons. While it is true that, in addition to autometamorphism produced by late magmatic fluids, some of the volcanic rocks are weathered, there is good evidence that this was a penecontemporaneous subaerial process, as in the Central Bohemian area kaolinite deposits (Havlena 1966). Nor is it likely that two different magnetization directions co-existed within a distance of 200 km, each consistent over a lateral distance of the order of 50 km (Table 2).

It is therefore suggested that the mean direc-

TABLE 3. SUMMARY OF PALEOMAGNETIC DATA

Region	Geographic co-ordinates		direction of magnetization					Virtual pole position		Ovals of confidence	
	lat.	long.	D	I	α_{95}	K	N	lat.	long.	dm	dp
Inner Sudetic Basin (Czech and Polish rocks)	50.6° N.	16.1° E.	195.1°	−1.8°	3.99°	17.61	76	38.69° N.	176.8° E.	3.99°	2.00°
Plzeň Basin	49.8° N.	13.3° E.	214.7°	2.1	5.09°	12.95	65	31.1° N.	151.7° E.	5.09°	2.55°
Kladno-Radovník Basin	50.2° N.	14.0° E.	211.7°	−5.4°	2.59°	42.02	73	35.5° N.	153.8° E.	2.60°	1.30°

α_{95} = circle of confidence at the 95% level.
K = measure of dispersion.
N = number of sites used.

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tions, that in the Inner Sudetic basin and those in the Plzeň and Kladno basins on the Central Bohemian massif, are original, and that the difference illustrated in Figure 10, which lies in declination rather than in inclination, is due to some subsequent tectonic effect. The obvious interpretation is that this tectonic effect was a relative rotation of the order of 17° between the two regions. If the virtual pole positions computed are plotted against other published virtual pole positions, it will be seen that they show a similar dispersion. It is therefore clear that tectonic effects, if such be the correct interpretation, may be as important a cause of dispersion of virtual pole positions as are secondary components of magnetization

section that the Westphalian and Stephanian directions found in the Inner Sudetic basin are in agreement with those from coeval rocks from other parts of Europe.

If this is so, then the Lower Paleozoic (Barrandian) measurements from the Bohemian massif (Bucha, 1961, 1965) must appear suspect. Any tectonic effect (rotational movement) must postdate the youngest of the rocks examined, the zone 4 red beds which are the highest part of the Stephanian. Younger beds on the Bohemian massif, unfortunately, are not known, and consequently an upper age limit cannot be placed upon the time when the proposed movement took place. The results of studies of less well-dated hydrothermal ores

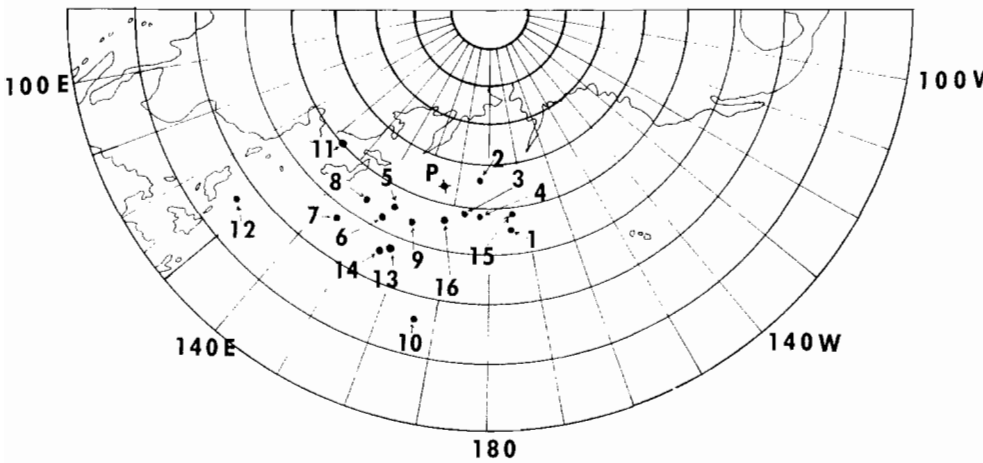


Figure 11. Virtual pole positions for various European Carboniferous rocks. Virtual poles are numbered as in Table 4. The data are mostly from Irving (1964). ⊕ P is the Permian mean virtual pole position; points numbered 1–7 European Westphalian and Stephanian results; points numbered 8–11 British Upper Carboniferous results; points numbered 12–14 British and other Lower Carboniferous results.

which have not been eliminated. The first approximation, of assuming uniformity of conditions over blocks of continental extent, must therefore give way to a more refined treatment in which sufficient attention is given to geological detail.

Given that the Permian directions of magnetization from both sides of the Sudetic basin are in accord with those of central and northern Europe and that the immediately preceding Stephanian and Westphalian of the Inner Sudetic basin give similar directions, then it must inevitably follow that the tectonically affected region was the Central Bohemian massif or part of it. It will be shown in the following

from Krušné Hory (Erzgebirge) in the north of Bohemia, from Freiberg in East Germany, and from the southern part of the Central Bohemian plateau (Krs, 1963, 1964; Hanuš and Krs, 1963) suggest that the tectonically affected region is quite extensive.

Such a conclusion is contrary to much of geological opinion, which regards the Bohemian massif as highly stable. The theory may perhaps be hazarded that it is due to just this strength and rigidity that a massif yields to applied (tectonic?) stresses by fracture and shear with which the rotation may be associated; in a mobile zone, however, deformation by folding and thrusting may occur. It is dif-

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TABLE 4. CARBONIFEROUS VIRTUAL POLE POSITIONS

Permian of Meso-Europe, mean pole position		43° N.	166° E.
Carboniferous			
1. France and Germany	Stephanian (A & B in Saar)	36° N.	175° W.
2. U.S.S.R.	Krasnotsvet clay Stephanian	47° N.	178° E.
3. . . .	Donbass combined "upper"	39° N.	173° E.
4. Poland and Č.S.S.R.	combined Westphalian B/C–Stephanian	38.7° N.	176.8° E.
5. . . .	Kladno basin Westphalian C–Stephanian	35.5° N.	153.8° E.
6. . . .	Plzeň basin Westphalian C–Stephanian	31.1° N.	151.7° E.
7. Great Britain	Keele beds combined ?Westphalian or low Stephanian	28° N.	144° E.
8. . . .	Derbyshire sediments combined Carb. Limestone-Millstone Grit	33° N.	148° E.
9. . . .	Lancashire sediments combined Carb. Limestone-Millstone Grit	34° N.	160° E.
10. . . .	Kinghorn Lavas Calciferous Sandstone Series	17° N.	167° E.
11. U.S.S.R.	Siberian "Lower Carboniferous"	40° N.	131° E.
12. . . .	Tikvin Tournaisian-Namurian	17° N.	127° E.
13. . . .	S. Urals Tournaisian	29° N.	159° E.
14. Great Britain	Midland sills and sediments*	28° N.	157° E.
15. . . .	Little Wenlock "lava" and sediment*	39° N.	172° W.
16. . . .	Great Whin Sill*	37° N.	169° E.

(*after* IRVING, 1964)
* Radiometric age data of Fitch and Miller (1964)

ficult to find geological evidence which can be brought to bear on the question of rotation. There are two indications known to the authors, one more probable than the other, of important transcurrent displacements which are in the same sense as the rotation here proposed would require. The first is reported by Pietzsch (1962) in that part of Saxony lying immediately to the west of the Elbe line. The Elbe line, whose importance can be first recognized in the Sudetic movements at the close of the Lower Carboniferous, parallels the Baltopodolian line which, according to Stille (1949), effectively forms the border of the ancient Fennosarmatian shield area. Pietzsch (1962, p. 795) indicates relative northward movement in Saxony immediately west of the Elbe line relative to the area to the east. Rather more hypothetical is Stille's (1949) suggestion that the Rhine line which appears to end against the Alpine front is displaced to the northwest, its continuation forming the Rhone-Saône trough. The line along which this displacement is said to occur is parallel to the Baltopodolian and Elbe lines (see Fig. 1).

The Relevance of the Sudetic and Bohemian Results to the Measurements Made on Other Carboniferous Rocks in Europe

The interpretation of the virtual pole positions from the Carboniferous rocks of Europe has long posed a problem. The numerous published virtual pole positions show a scatter so great that Irving (1964) found it convenient to treat them in two separate groups. On the other hand, Permian rocks from the same region provided virtual pole positions whose dispersion was surprisingly small. The youngest Carboniferous results, those from the Stephanian of western Europe (Nairn, 1960), seemed to fall close to the Permian directions and so be distinguished from the older Carboniferous which, however, contained measurements on rocks up to Westphalian age. Everitt and Belshé (1960) attempted to attribute this dispersion to a secondary magnetization imprinted on the original, for in most of the earlier measurements no magnetic stability or cleaning procedures had been carried out. The clear association of the anomalous magnetic results found in Permian rocks south of the Alpine front with tectonic activity is not to be seen immediately in Carboniferous rocks. It is therefore of cardinal importance to consider the relevance of the Czech and Polish results described here to the more general problem of the Carboniferous of Europe as a whole.

The principal results listed in Table 4 with which the Sudetic virtual pole position will be compared are abstracted from Irving (1964). The first thing to be observed is that the results from the Inner Sudetic basin agree closely with the Stephanian determinations from the Massif Central and the Saar (Nairn, 1960), the Soviet Union (Krasnosvet and Donbas, *in* Irving, 1964) and Great Britain. The British

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results are known from radiometric dating of some "Upper Carboniferous" igneous rocks (Fitch and Miller, 1964). This radiometric work has shown that certain igneous rocks have ages grouping at about 295 m. y., and are therefore within the Stephanian, although the uncertainties of the age determinations exceed its duration. Of these the best known is the Great Whin Sill, an intrusion hitherto regarded on geological grounds as dating from about the Carboniferous-Permian boundary. From the results of a paleomagnetic study, Creer, Irving, and Nairn (1959) concluded that the directions of magnetization of the intrusion were not significantly different from those of the Permian. It further appears that the Little Wenlock "Lava" formerly regarded as Lower Carboniferous has a similar age. Their virtual pole positions are shown in Figure 11. Three of the Upper Carboniferous intrusions mentioned by Everitt and Belshé (1960) turn out to be Permian in age, a fourth has a Stephanian age, and the remaining two are undated. Of these results, stability tests have only been carried out on the igneous rocks.

The remaining Upper Carboniferous rocks from Great Britain, from the Lancashire coal field, from the Derbyshire coal field, and from the Keele beds of the Midlands show varying degrees of dispersion, and it would now seem necessary to re-examine these areas to see whether there is a significant secondary component of magnetization, or whether here too tectonic effects are involved. The major earth movements in Britain occurred during the lower part of the Stephanian, and there is evidence in the case of the Whin Sill that intrusion postdated the tectonic activity (Collinson and Nairn, 1960).

At the moment little can be said about the scatter of the results from the Lower Carboniferous, except to remark that the only reliable Lower Carboniferous measurements, in the sense that they were subjected to magnetic cleaning, and were sufficient in number and time coverage, are measurements of the Kinghorn lavas.

CONCLUSIONS

The 17° difference between the mean direction of magnetization of the Upper Carboniferous rocks of the Inner Sudetic basin and that of the basins on the Bohemian platform, the Kladno-Raknovík and Plzeň basins, is significant. Stability tests have shown that this difference cannot be attributed to some residual secondary component of magnetization. Neither is it associated with a particular lithology, in view of the variety tested. Similarly, there is no reason to suppose it to be due to different geological situations resulting in the entire remagnetization of one section. The geology of both regions, because of the economic importance of the beds, is known in better detail than is the case in many studies. As the concept of two co-existing poles seems unreasonable, the difference must be attributed to some tectonic effect.

From the paleomagnetic point of view, this detailed study of two regions less than 200 km apart in the heart of Europe commends a certain caution in the interpretation of virtual pole positions. It suggests that the interpretation of paleomagnetic directions should include consideration of the tectonic environment. If the dispersion of virtual pole positions is a reflection of different tectonic histories even within a limited area, then a more refined basis for intercontinental comparisons of paleomagnetic results than simple continental blocks is required.

When the mean directions of the two regions, the Inner Sudetic basin and the Bohemian platform, are compared with the results from coeval rocks from other parts of Europe, it is the measurements from the Bohemian platform which are divergent. In consequence, if a tectonic effect caused the divergence, it must have affected the Bohemian platform. No precise date can be proposed for the tectonic activity, in the absence of younger beds on the massif. The simplest interpretation, geometrically, if not geologically, involves a relative rotation of the Bohemian massif. This is a somewhat disturbing conclusion, in view of the generally accepted views on the stability of cratonic blocks. Although consistent with the megatectonic views of Stille (1949), any such displacement obviously involves spatial problems.

The virtual pole position for the central Bohemian region, uncorrected, lies in the same general region as some of the British Carboniferous results from the coal field areas. It suggests that reinvestigation, with appropriate stability tests, is advisable in order to establish whether or not significant tectonic effects are involved. It also suggests that if the tectonic interpretation is correct, the lower Paleozoic Barrandian results from Bohemia are suspect and will require correction.

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